WEEK 5

Reinforcement Learning: The Idea

From deterministic rules to adaptive agents — a physicist's perspective

Bellman Equation

Q-Learning

Simulated Annealing

The Big Conceptual Shift

What changes when agents learn?

WEEKS 1–4

Fixed Rules

- Agents follow predefined transition rules
- No memory of past outcomes
- Behavior never changes
- Emergence from structure, not adaptation

Conway, Schelling, Sand Piles →
rules are the model

WEEK 5 →

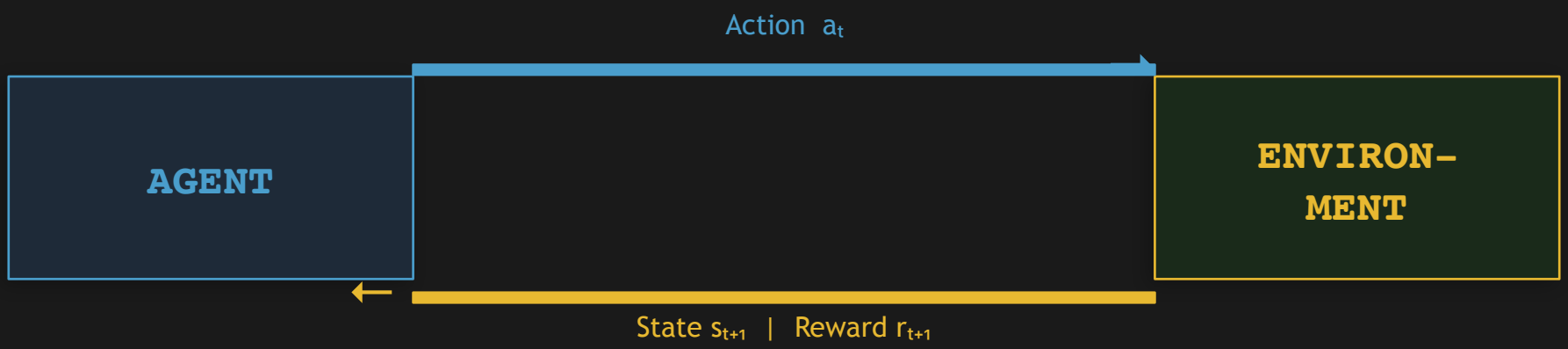
Learning Agents

- Agents update behavior from experience
- Optimize a long-term objective (reward)
- Rules emerge from interaction
- Trade-off: exploit known vs explore unknown

Emergence now from optimization,
not fixed structure

The Reinforcement Learning Framework

Four ingredients are all you need



State s_t
Complete description of the world at time t

Action a_t
Choice the agent makes given the current state

Reward r_t
Scalar feedback signal from the environment

Policy $\pi(s)$
Mapping from states to actions — the strategy

The Physics Perspective

RL as optimization on an energy landscape

Statistical Physics

System minimizes free energy F

$F = U - TS$

U = internal energy (cost)
T = temperature (randomness)
S = entropy (disorder)

High T → random exploration of states
Low T → system settles in minimum

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ANALOGY

Reinforcement Learning

Agent maximizes cumulative reward V

$V(s) = E[\Sigma \gamma^t r_t]$

V(s) = value function (−energy)
γ = discount factor (horizon)
r_t = reward at time t

High ε → random exploration of actions
Low ε → exploit best known action

The Bellman Equation — Derivation

Step 1: Define the value function

Definition: Value Function $V^\pi(s)$

$$V^\pi(s) = E_\pi [r_0 + \gamma r_1 + \gamma^2 r_2 + \dots \mid s_0=s] = E_\pi [\Sigma_{t=0}^{\infty} \gamma^t r_t \mid s_0=s]$$

$\gamma \in [0,1)$ is the discount factor — future rewards matter less. $\gamma \to 0$: myopic agent. $\gamma \to 1$: far-sighted agent.

Key Observation: Recursion

$$V^\pi(s) = E_\pi [r_0 + \gamma (r_1 + \gamma r_2 + \dots) \mid s_0=s] = E_\pi [r_0 + \gamma V^\pi(s_1) \mid s_0=s]$$

The Bellman Equation — Full Form

A self-consistent equation for the value function

Bellman Expectation Equation

$$V^{\pi}(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s,a) \left[R(s,a,s') + \gamma V^{\pi}(s') \right]$$

$\pi(a|s)$

Probability of taking action a in state s under policy π

$P(s'|s,a)$

Transition probability: reaching s' from s via action a (environment dynamics)

$R(s,a,s')$

Immediate reward received when transitioning from s to s' via a

$\gamma V^{\pi}(s')$

Discounted value of the next state — the recursive future contribution

Physics note: This is a fixed-point equation — like the self-consistency condition in mean-field theory.

From V to Q — Action-Value Function

The quantity an agent can actually optimize

Definition: Q-function (action-value function)

$$Q^{\pi}(s,a) = E_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0=s,\ a_0=a \right] = R(s,a) + \gamma \sum_{s'} P(s'|s,a)\, V^{\pi}(s')$$

Bellman Optimality Equation

$$Q^*(s,a) = R(s,a) + \gamma \sum_{s'} P(s'|s,a) \max_{a'} Q^*(s',a')$$

Optimal policy: $\pi^*(s) = \text{argmax}_a\ Q^*(s,a)$

Key: Q* contains everything needed to act optimally — no model of the environment required

Q-Learning: The Algorithm

Model-free learning via temporal difference updates

Q-Learning Update Rule

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

α

Learning rate — how much we trust new information

r + γ max Q(s',a')

TD target: immediate reward plus discounted best future value

TD error δ

r + γ max Q(s',a') − Q(s,a) = prediction error to minimize

Initialize Q(s,a) = 0 ∀ s,a → Loop: observe s, choose a (ε-greedy), get r and s', update Q, set s ← s'

Exploration vs. Exploitation

The temperature analogy from statistical physics

ε-Greedy Policy

With probability ε :
choose a random action (explore)

With probability 1−ε :
choose argmax_a Q(s,a) (exploit)

Annealing schedule:
$\varepsilon(t) = \varepsilon_{min} + (\varepsilon_{max} - \varepsilon_{min})\, e^{-t/\tau}$

τ is the cooling timescale — exactly as in simulated annealing

Simulated Annealing

Physical metal cooling:

High T → atoms explore widely
Low T → atoms settle in minimum

RL agent cooling:

High ε → agent tries random actions
Low ε → agent exploits learned Q

Too fast cooling → local minimum trap
Too slow cooling → wasted computation

GridWorld: Our Minimal Test Environment

The hydrogen atom of reinforcement learning

GridWorld Rules

- **A** — Agent starts at S. Actions: Up/Down/Left/Right
- **G** — Goal cell: reward = +1. Episode ends on arrival
- ■ — Wall cells: blocked. Agent stays in current state
- • — Empty cell: reward = −0.01 (encourages efficiency)

Q-table: 25 states × 4 actions = 100 entries. Initialized to zero, updated by Bellman rule.

Q-Learning in Action: One Update Step

Tracing the Bellman update through a concrete example

Scenario: Agent in state s=(3,2), takes action a=RIGHT, arrives in s'=(3,3), receives r=−0.01

$\alpha = 0.1$	$\gamma = 0.9$	$Q(s,a) = 0.0$	$\max_{a'} Q(s',a') = 0.5$

Step 1: Compute TD target: $\text{TD target} = r + \gamma \max_{a'} Q(s',a') = -0.01 + 0.9 \times 0.5 = 0.44$

Step 2: Compute TD error δ: $\delta = \text{TD target} - Q(s,a) = 0.44 - 0.0 = 0.44$

Step 3: Update Q-value: $Q(s,a) \leftarrow Q(s,a) + \alpha \times \delta = 0.0 + 0.1 \times 0.44 = 0.044$

After many episodes: Q-values converge to Q — the Bellman fixed point — regardless of initialization.*

Convergence: A Physicist's Reading

Q-learning as stochastic fixed-point iteration

Convergence Theorem

Q-learning converges to Q* if:

1. Every (s,a) pair visited infinitely often
2. Learning rate α satisfies:
   $\sum \alpha(t) = \infty$
   $\sum \alpha(t)^2 < \infty$

Same Robbins-Monro conditions as stochastic gradient descent

The Physics Reading

Q* is a fixed point:

$T[Q](s,a) = R + \gamma \max Q(s',a')$

T is a contraction mapping ($\gamma < 1$)
→ unique fixed point guaranteed

Analogous to:
Renormalization group flow → stable fixed points as RG attractors

Week 5 — Summary

What we built, what comes next

RL Framework

State, action, reward, policy — the four ingredients of any learning agent

Bellman Equation

Self-consistent fixed-point equation connecting current and future value

Q-Learning

Model-free TD algorithm provably convergent to the optimal policy

Temperature Analogy

ε-annealing mirrors simulated annealing: cool slowly to avoid local optima

Week 6 → Gymnasium: we run CartPole and FrozenLake, train them with Stable-Baselines3, and analyze the learned policies as physicists.